MoE Parallel Folding

Give MoE layers their own parallel mapping

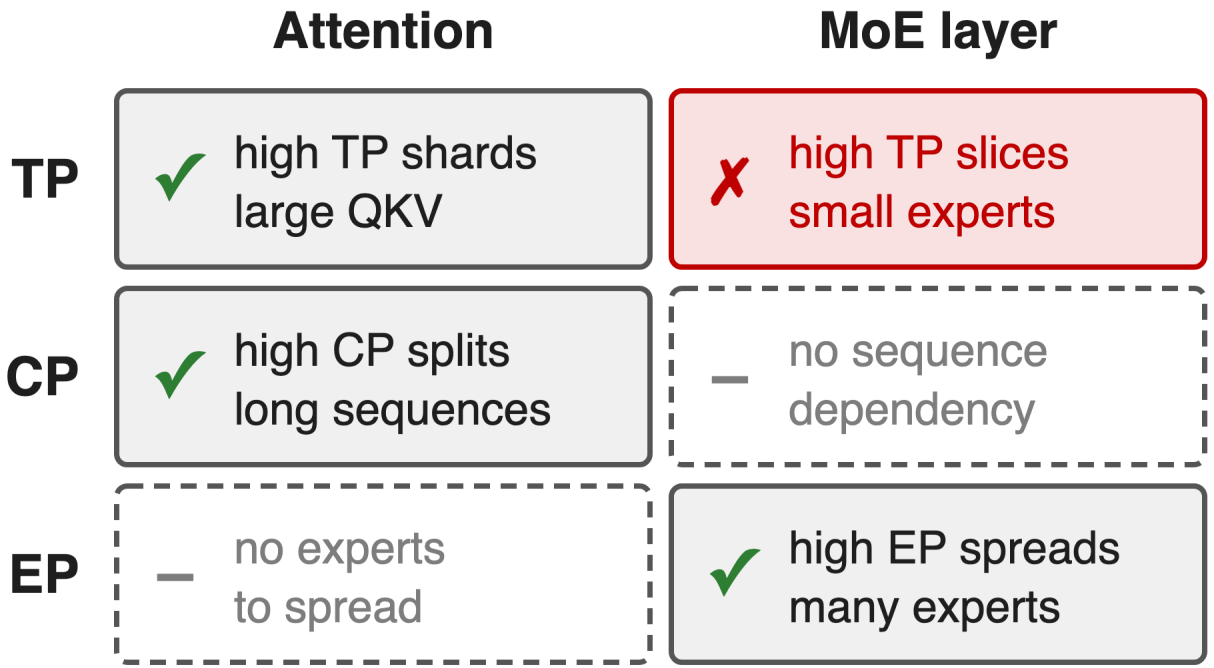
NVIDIA · Megatron-Core MoE

Explainer deck · Industrial Pretraining Architecture & Infra Tech Landscape (2026)

Parallelism Strategy · arxiv.org/abs/2603.07685

Attention and experts want opposite parallelism

- DeepSeek-V3: **685B** params, **37B** active (**18:1**)
- Experts must spread over many GPUs
- Attention wants TP and CP; experts want EP



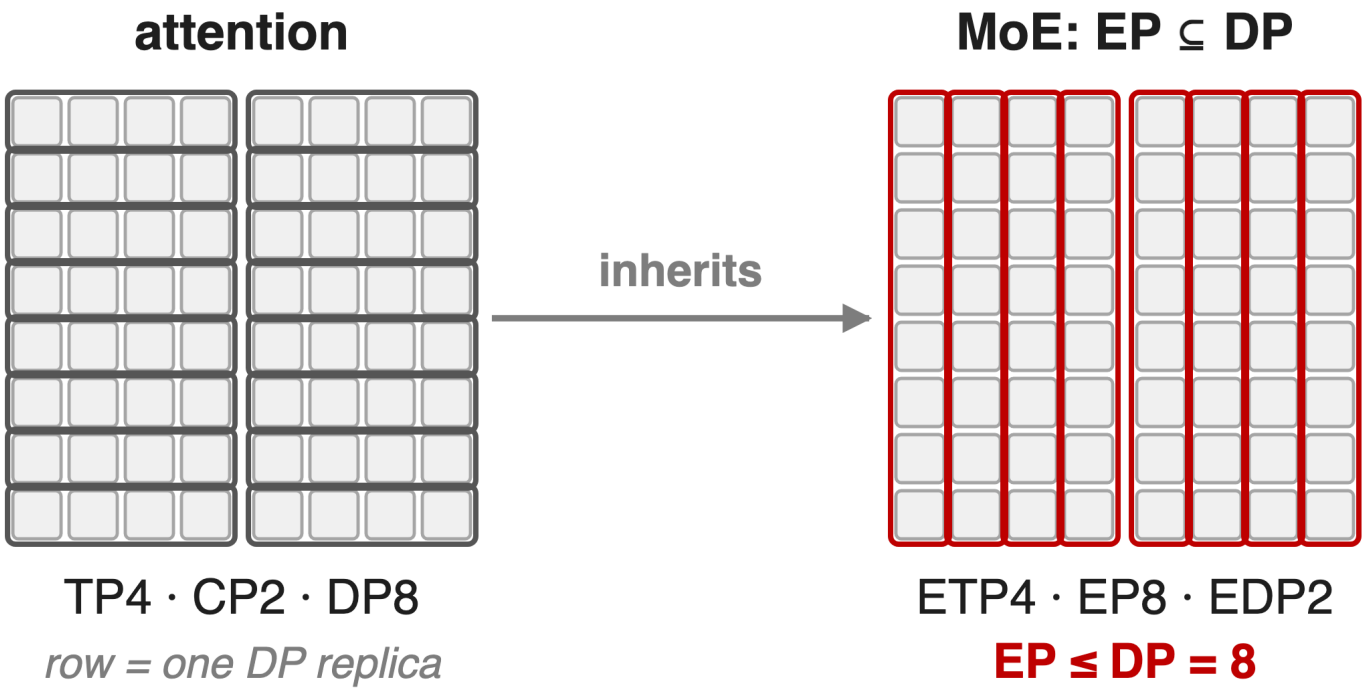
Fine-grained MoE stalls at 17% MFU

17.1%

MFU of the best Megatron-Core configuration for fine-grained Mixtral-8x22B-G8T8 on 128 H100 when attention and MoE share one mapping

MoE Parallel Folding, arXiv 2504.14960, Table 1 (BF16, global batch 256)

Sharing one mapping caps EP at DP



256 GPUs as TP4 · CP2 · DP8 · PP4: EP can never exceed the 8 DP replicas

Neither knob setting fixes both layers

- **Raise TP for attention**
- Experts slice into thin shards
- Fine-grained: **> 70%** of MoE time is communication off-node

- **Raise EP inside DP**
- CP8 + EP8 needs **64 GPUs**
- All-to-all leaves NVLink: **5–10×** less bandwidth

Root cause: one parallel mapping is forced on two layer types

Insight: experts see a bag of tokens

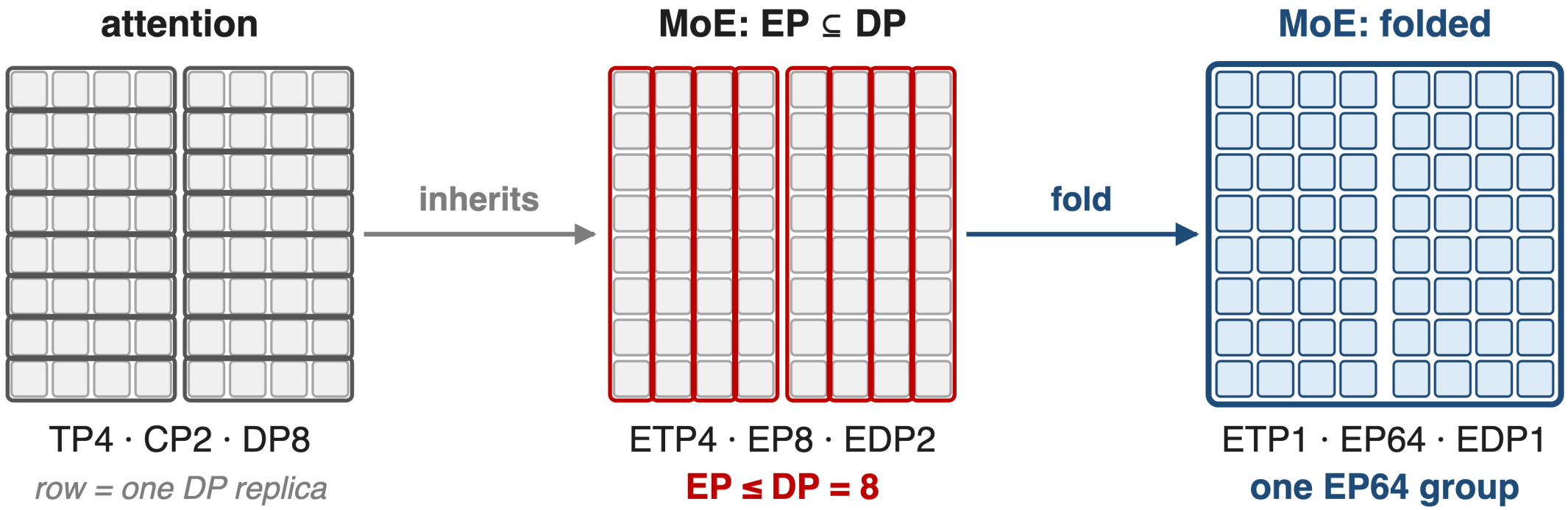
- Attention splits tokens by batch (DP) and sequence (CP, TP-SP)
- An expert computes every token independently

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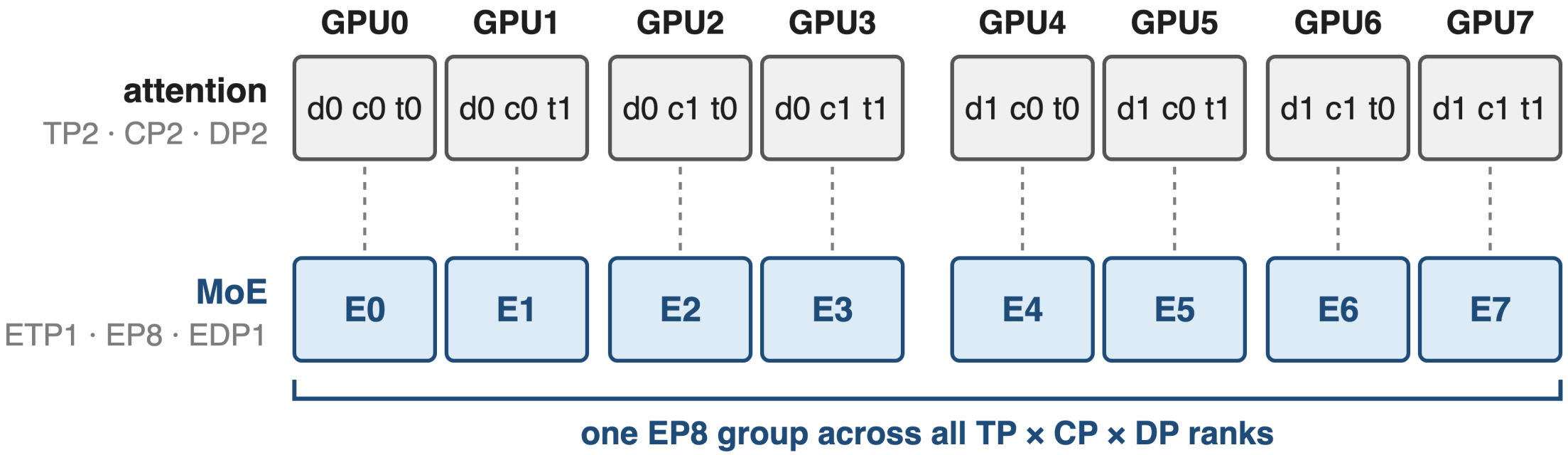
So the MoE layer can re-factor the same GPUs freely; only pipeline stages must match

Folding gives MoE its own grid on the same GPUs



Same 64 GPUs: EP grows from 8 to 64 (8×) while attention keeps TP4 · CP2

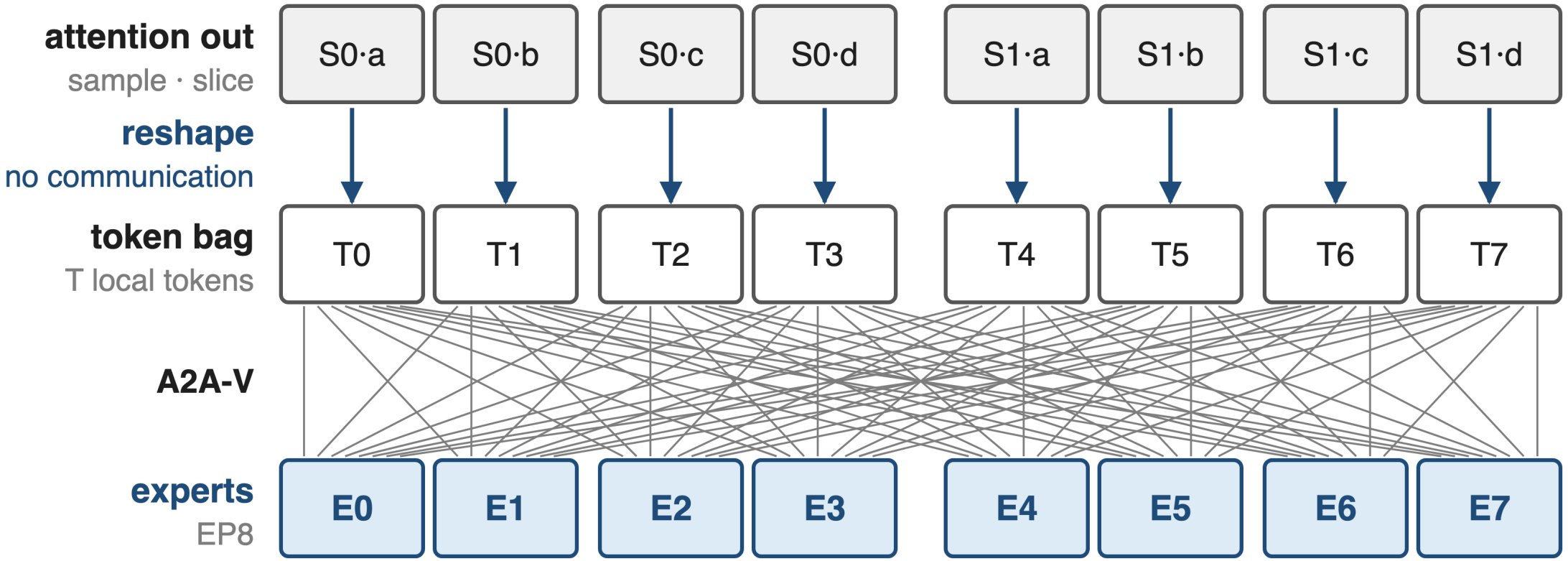
❶ Two rank grids tile the same GPUs



rank order tp-cp-ep-dp-pp: TP fastest, PP outermost and shared

Attention groups over TP×CP×DP×PP, MoE groups over ETP×EP×EDP×PP; Megatron-Core asserts identical PP groups

❷ A reshape, not a collective, joins the grids



If ETP > 1 the dispatcher adds AllGather-V / ReduceScatter-V inside each ETP group

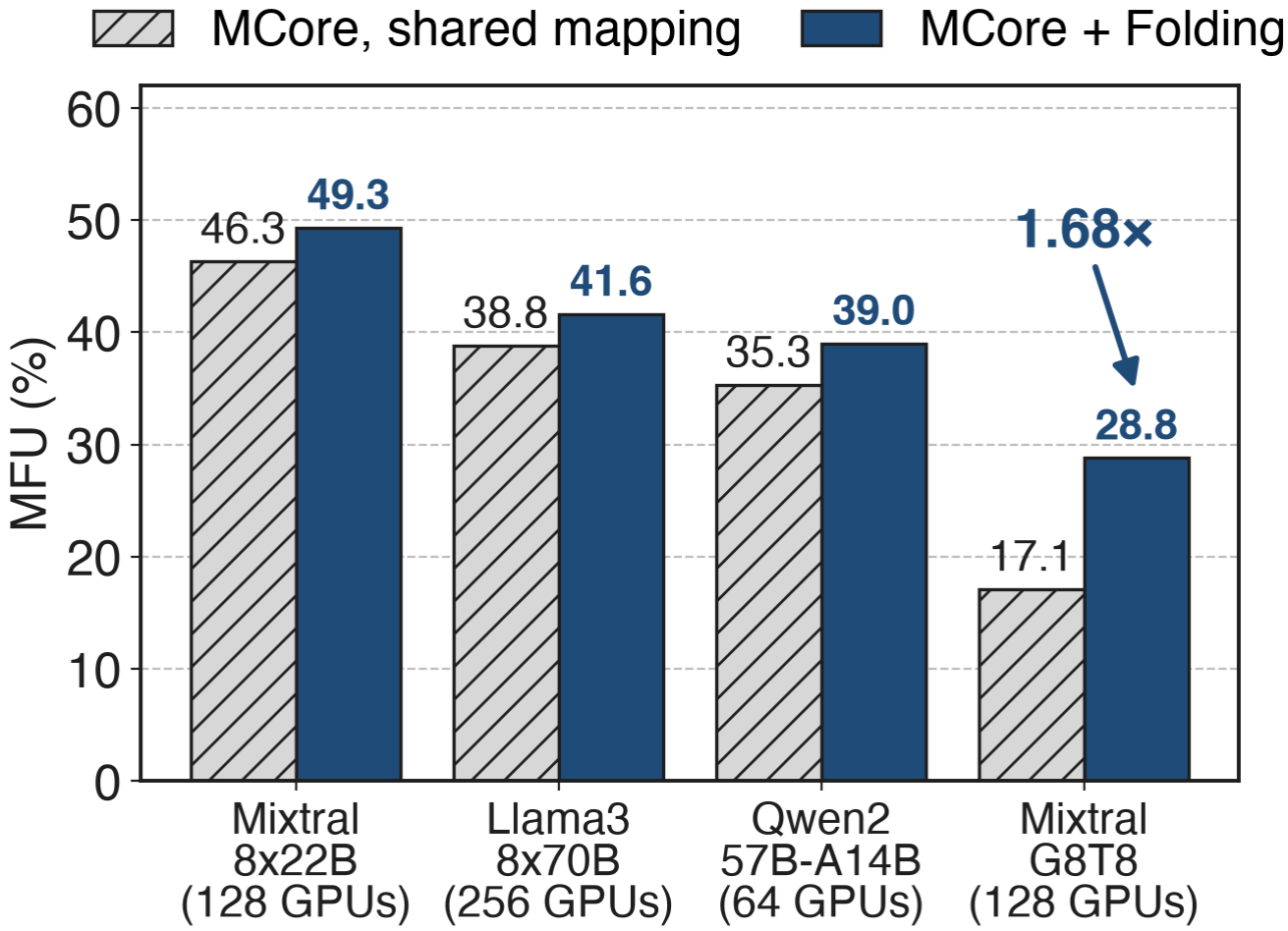
❸ Expert gradients reduce over their own group

- Dense gradients reduce over **DP × CP**
- Expert gradients reduce over **EDP**, scaled by EDP / DP
- Separate optimizers; expert states sharded over EDP

Check: Mixtral 8x7B loss matches unfolded Megatron-Core over 40B tokens

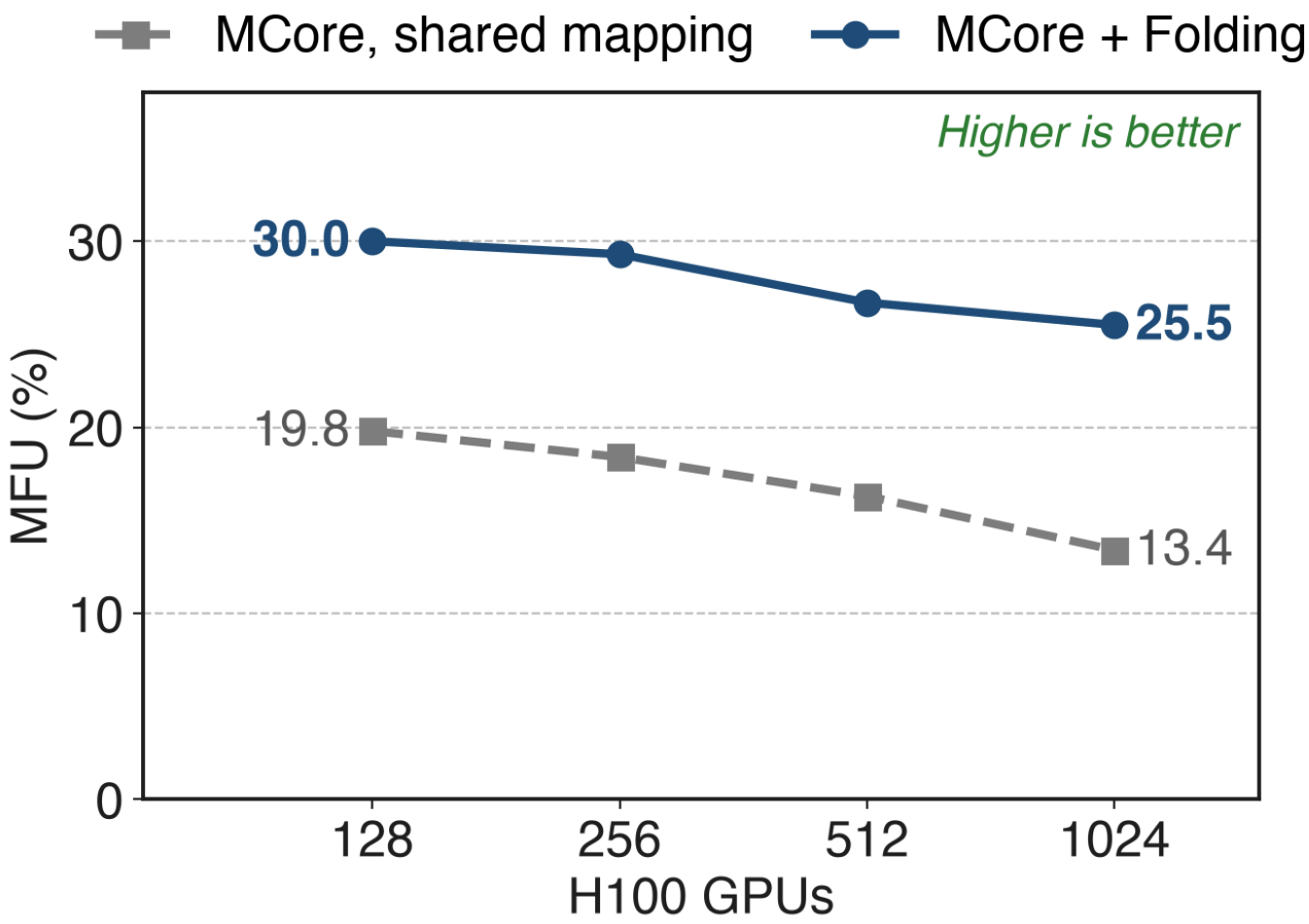
Fine-grained MoE: 17.1% → 28.8% MFU

- **1.68×** on G8T8 (derived)
- +2.8 to +3.7 points on three other MoEs
- G8T8 folded: attention TP4; experts EP8, ETP1



At 1,024 GPUs: 25.5% vs 13.4% MFU

- Fine-grained G8T8, global batch 1024
- The shared mapping loses more as GPUs grow
- FP8 Mixtral-8x22B: **631.7** vs 575.1 TFLOPS



DeepSeek-V3 production recipes use folding

- 1,024 H100: attention TP2 · PP8; experts EP64 · ETP1: **368** TFLOPS/GPU
- 256 GB200: TP1 · PP4 · EP64: **1,048** TFLOPS/GPU
- EP64 puts exactly **4** of 256 experts on each GPU

The cost: one PP, explicit ETP, NVLink-sized EP

- Pipeline stages must match in both grids
- ETP defaults to TP: set `--expert-tensor-parallel-size 1`
- EP past NVLink still pays cross-node all-to-all
- Token dropping decided per sub-sequence

Summary

- One shared mapping caps EP at DP and slices small experts
- **Insight**: experts see a bag of tokens, so re-factor the same GPUs
- Same 256 GPUs: EP **8** → **64**
- Fine-grained MFU **17.1%** → **28.8%**; **25.5%** vs 13.4% at 1,024 GPUs

https://arxiv.org/abs/2603.07685